Qualifying Exam in Microeconomics
Arizona State University
June 2016

There are four questions and you have four hours to complete the exam.

1. There is a pure exchange economy with three consumers indexed by $i = 1, 2, 3$. The economy has three goods x, y, and z. Agent i's consumption vector is denoted by (x_i, y_i, z_i). Consumer preferences and endowments are

$$U_1(x_1, y_1, z_1) = x_1(y_1 + z_1) \qquad \omega_1 = (1, 0, 0)$$
$$U_2(x_2, y_2, z_2) = x_2 y_2 \qquad \omega_2 = (0, 1, 0)$$
$$U_3(x_3, y_3, z_3) = x_3 z_3 \qquad \omega_3 = (0, 0, 1)$$

As an example, consumer 2's endowment consists of one unit of good y.

(a) Find a Walrasian equilibrium under the assumption that good z cannot be traded, that is to say, there is no market for good z.
Is the equilibrium Pareto optimal? ▷ constraint to not trading z?

(b) Find a Walrasian equilibrium under the assumption that all goods can be traded.
Is the equilibrium Pareto optimal?

(c) Can the equilibria that you found be Pareto ranked?

2. A family has one parent and one child in an economy with a single consumption good. The parent is benevolent, caring not only for its own consumption, c_0, but also for the child's consumption, c_1. The parent's preferences over family consumption $c = (c_0, c_1)$ in $\mathbb{R}^2_+$ are represented by $U(c) = u_0(c_0) + u_1(c_1)$, where each u_i is *strictly increasing*, *strictly concave*, *differentiable*, and satisfies the Inada condition $u_i'(0) = \infty$.

The child chooses an action that determines the supply of the consumption good available to the household, and the parent chooses a family consumption profile (a "bequest").

Specifically, consider this game. First the child choses an action in a nonempty, compact real interval A. The supply of the good available to the family is a continuous function $\omega : A \to \mathbb{R}_{++}$ of the action a. The parent observes the the action, then chooses a consumption profile $(c_0, c_1) \geq 0$ subject to the constraint that $c_0 + c_1 \leq \omega(a)$. Any cost of the action is already included in ω, so the child just cares about its final consumption, c_1, the higher the better.

(a) Identify the strategy set for each player.

(b) *Evaluate*: "In *any* subgame perfect Nash equilibrium (SPNE), the child's action maximizes the household's supply of the good on A." Prove if true, give a counterexample or explanation if false.

(c) What role, if any, does the joint assumption of additive separability and strict concavity of the parent's preference representation play?

(d) Does the parent have a weakly dominant strategy in this game? Explain.

3. Consider the following two player game.

The game is as follows. There are two players indexed by $i = I, II$. Player i observes her private information t_i in $\{h, l\}$, each realization has probability 1/2. The private information variables are independently distributed. After observing their private information, players simultaneously and independently choose actions. Player i chooses an action a_i in $\{x, y\}$.

Payoffs are as follows. If a player chooses x and the other player chooses y, the player that chose x gets a payoff 8 and the other one a payoff of 0.

If both players choose y both get a payoff of 0.

If both players choose x and both players are type h, then both get a payoff of -12 each.

If both players choose x and both players are type l, then both get a payoff of 4 each.

If both players choose x and one of them is type h while the other one is type l, then the type-h player gets a payoff of -12 and the type-l player gets 4.

(a) Write the extensive form of the game described.

(b) Find a pure strategy Bayesian Nash equilibrium. Is there another pure strategy Bayesian Nash equilibrium?

→ normal form!! →Better way??

4. A firm produces one output q using two inputs, labor ℓ and capital k. Labor can be any nonnegative real number but capital can only take on values in a *finite* set $K \subset \mathbb{R}_+$. Its technology is described by a production function $f : \mathbb{R}_+ \times K \to \mathbb{R}_+$. The function f is *strictly increasing* on $\mathbb{R}_+ \times K$; and, for each $k \in K$, $f(0, k) = 0$, and the function $f(\cdot, k)$ is *unbounded above* and *continuously differentiable* with $f_\ell(\ell, k) > 0$. The unit price of labor is w and the unit price of capital is r, both positive. Consider the firm's *cost minimization* problem for $q > 0$:

→ not a lattice

$$\min_{(\ell,k)}\{w\ell + rk\} \quad \text{subject to} \quad (\ell, k) \in \mathbb{R}_+ \times K \quad \text{and} \quad f(\ell, k) \geq q. \tag{1}$$

(a) Prove the *law of input demand* that an increase in r *does not increase* the firm's choice of capital: if $r_1 > r_0 > 0$ and (ℓ_i, k_i) solves (1) at (q, w, r_i), then $k_1 \leq k_0$.

(b) Define a function $g : \mathbb{R}_+ \times K \to \mathbb{R}_+$ by the equation $g(q, k) = \min\{\ell \in \mathbb{R}_+ \mid f(\ell, k) \geq q\}$. Explain in words what this function is. Confirm that it is differentiable in q.

(c) Find a condition *directly* on the function g for the *normality* of capital in the sense that an increase in q *does not decrease* the firm's capital choice: if $q_1 > q_0 > 0$ and (ℓ_i, k_i) solves (1) at (q_i, w, r), then $k_1 \geq k_0$. Justify your answer.

(d) Identify properties of f which imply the condition you found in part (c).